

O-RAN VIP Suite — Red-Team of the Signoff

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Question under test: **what does 1522/1522 (now 1523/1523) NOT prove?** · Every finding verified against source, file:line cited.

Bottom line. 1523/1523 proves the generated testbench is **internally consistent and runs clean on Verilator**: the spec→stimulus→model→checker→coverage pipeline executes end-to-end and reproducibly, each header codec's bit-layout is self-consistent, and the infrastructure (gates, adjudication, coverage merge, reporting) works. It does **not** prove that a real RU design is correct (there is no RTL design under test), that the model matches the authoritative O-RAN spec (requirements are DERIVED and self-checked), that coverage is adequate (bins are self-derived; covergroup % uncomputed), that the assertions check protocol behaviour (procedural counters / one vacuous SVA), or that the checkers can catch a bug they did not themselves plant (zero mutation testing; zero bugs ever found). **This is a VIP/model integrity result, not a DUT-signoff result.**

What 1523/1523 legitimately DOES prove

- The full generate→compile→run→adjudicate→coverage→report flow executes end-to-end for all 13 components + RAL, deterministically (seed-locked) on Verilator 5.020/5.050.
- Each 64-bit header codec is bit-layout self-consistent: `SV pack` matches the Python golden pack, and `unpack` inverts `pack`.
- The negative path's legality function fires on injected-illegal fields (`EXPECTED_FAILURE_DETECTED`), and the adjudicator derives `STATUS` from executed log evidence, not inference.
- The RAL exercises the standard `uvm_reg` flow (map/adaptor/predictor, front-door RW, RO write-protect) against a memory model, with 0 mismatches.
- As a **reusable VIP scaffold and methodology harness**, the suite is real, working engineering.

Findings — ranked by severity

BLOCKER RT-001 · There is no RTL design under test

CLAIM

The "block under test" is a purely combinational pack/unpack codec — no clock, no state, no protocol behaviour. Signoff therefore cannot assert design correctness.

EVIDENCE

`components/<slug>/rtl/<slug>_codec.sv` — 13/13 codecs contain zero clocked logic (no `posedge` / `always_ff`):

```
module uplane_codec (...); assign wire_hdr = uplane_pack(in_hdr); assign out_hdr = uplane_unpack(rx_wire); endmodule
```

WHAT 1523 DOES NOT PROVE

Nothing about an RU/DU implementation, FSMs, timing, back-pressure, or any sequential protocol behaviour. The pass count measures a model, not a design.

FIX BEFORE SIGNOFF

Re-label the deliverable as **VIP + reference-model signoff**, OR bind the VIP to a real RTL DUT (or a behavioural reference with injected faults) and re-run. Do not present as design signoff.

BLOCKER RT-002 · Self-referential pipeline — one source of truth feeds stimulus, DUT, checker AND coverage

CLAIM

`spec.json` generates the constraint set, the codec, the `is_legal / first_violation` checker, and the coverage bins. Positive-pass is near-tautological: Z3 emits a legal header; the same spec's checker agrees.

EVIDENCE

`gen/gen_stim.py` `py_is_legal()` is described in-file as "mirror of SV `is_legal`"; both derive from the same spec. `uplane_first_violation()` encodes the identical bounds the solver was given.

WHAT 1523 DOES NOT PROVE

Independence of stimulus and checking. A conformant generator and a conformant checker built from the same text cannot disagree except on an implementation typo — so agreement is expected, not evidence of correctness.

FIX BEFORE SIGNOFF

Introduce an **independent** golden (different author/derivation, or the authoritative spec's own vectors) so the checker is not a restatement of the generator.

HIGH RT-003 · Checker sensitivity is unproven — zero mutation testing, zero bugs ever caught

CLAIM

The testbench has never caught a defect it did not itself plant, so its power to detect a real, independent bug is unknown.

EVIDENCE

`grep -rIn "BUG-[0-9]"` across `regression/ lane2/ components/*/sim_out/` → empty. No BUG-### in any ledger.

WHAT 1523 DOES NOT PROVE

That a 100% pass rate distinguishes a correct model from a checker that catches nothing. Both produce all-green.

FIX BEFORE SIGNOFF

Run **mutation / fault-injection testing**: perturb the codec (swap a field slice, off-by-one a bound, drop a cross term) and require the scoreboard to FAIL. Report kill-rate as the real quality metric.

HIGH RT-004 · The strongest check is a mathematical tautology

CLAIM

Round-trip (`unpack(pack(h)) != h`) can only fire if pack/unpack aren't inverses — it validates the TB's own bit-slicing. Pack-vs-golden compares two implementations of the *same* layout.

EVIDENCE

`tb/uplane_tb_top.sv`: `m_wire=pack(m_in); m_out=unpack(m_wire); if(m_out!=m_in) ROUNDTrip err; if(m_wire!=r_gold) PACK err`. Golden `r_gold` is the Python pack of the same fields.

WHAT 1523 DOES NOT PROVE

Any protocol-semantic property (field meaning, ordering, timing). It proves the layout is invertible and two encoders agree.

FIX BEFORE SIGNOFF

Add checks against **spec-sourced expected values** (decode known-good captures / vendor vectors), not against the suite's own encoder.

HIGH RT-005 · "Assertions" are procedural counters; the only concurrent SVA is vacuous

CLAIM

Lane-1 `SVA_EXERCISED,...` is a count of function calls, not temporal SVA. The one real concurrent-SVA interface (`ecpri_if.sv`) is transaction-decoupled, so `valid` never asserts and the properties are vacuous.

EVIDENCE

`tb/*_tb_top.sv`: `SVA_EXERCISED,legality=%0d,roundtrip=%0d,pack=%0d` printed from `m_ex*` counters. `ecpri_if.sv` properties gate on `valid`; driver publishes to the analysis port and never drives `vif` (see `VCS_PORTABILITY_NOTES`).

WHAT 1523 DOES NOT PROVE

That any temporal/ordering/handshake invariant was actually evaluated. Vacuous assertions pass unconditionally.

FIX BEFORE SIGNOFF

Wire the interface (NBA @posedge pattern already documented), add a **cover** per property and fail on vacuity, and add monotonic-`seq_id` stimulus so `p_seq_inc` is non-vacuous.

MED RT-006 · Coverage measures the generator hitting its own bins

CLAIM

100% functional is over bins the same `spec.json` defined; it proves the generator reached its own targets, not that the protocol space is adequately sampled. Native covergroup % was never computed on Verilator.

EVIDENCE

`suite_regression_full.json`: `func_holes total = 0` over self-derived COV-### bins. `lane2b_report.json`: "Verilator get_coverage not collected"; `get_coverage()`=0.

WHAT 1523 DOES NOT PROVE

Coverage *adequacy* (are the bins the right ones?), cross-coverage depth, or the true covergroup number.

FIX BEFORE SIGNOFF

Have coverage bins reviewed against the authoritative spec by someone other than the bin author; compute real covergroup % on VCS/Questa/Xcelium.

MED RT-007 · Code-coverage denominator is trivial (~15 lines); toggle ~44-61%

CLAIM

100% statement/branch is over ~15 lines of combinational codec per component — a denominator small enough that 100% carries little assurance. Toggle sits at ~44-61%.

EVIDENCE

`code_cov.json`: line total = 15 per component; toggle e.g. `cpri_eth` 43.75%, `uplane` 60.94%.

WHAT 1523 DOES NOT PROVE

Structural exercise of any non-trivial logic. There is little logic to cover.

FIX BEFORE SIGNOFF

State code coverage against the **real DUT** once bound; treat the current number as model-integrity only.

MED RT-008 · Requirements DERIVED, authoritative spec revision unpinned

CLAIM

Field bounds, enums and cross-constraints are DERIVED, not reconciled to a pinned O-RAN.WG4.CUS revision. Model-to-spec fidelity is unmeasured.

EVIDENCE

GAP-ORAN-001 (open) in tracker/VRD; GATE-0 marked DERIVED.

WHAT 1523 DOES NOT PROVE

That the modelled legal space equals the spec's legal space. A wrong bound would be enforced consistently by generator and checker and still show 100%.

FIX BEFORE SIGNOFF

Pin CUS R004-v16 (or current), diff every field/enum/cross against clause text, resolve GAP-ORAN-001.

MED-LOW RT-009 · One simulator actually ran; commercial path never compiled

CLAIM

All PASS evidence is Verilator. VCS/Questa/Xcelium are NOT_RUN; the double-UVM defect (now fixed) means the native-UVM path had never compiled on a commercial sim.

EVIDENCE

portability matrix (NOT_RUN); VCS_PORTABILITY_NOTES (double-UVM root cause, run pending).

WHAT 1523 DOES NOT PROVE

Simulator-independence, or that the UVM elaborates on a vendor tool.

FIX BEFORE SIGNOFF

Run `./run_commercial.sh vcs all 1` on a licensed host; record real logs; close the portability row.

LOW RT-010 · Header-only scope; behavioural protocol layers excluded

CLAIM

Timing servo (T12/T34), beam-weight math, compression SNR, PTP servo/holdover, concatenation reassembly, per-symbol ordering are explicitly out of scope.

EVIDENCE

FUNCTIONAL_DESIGN §5 "Out of Scope".

WHAT 1523 DOES NOT PROVE

Any dynamic/behavioural conformance — only static header field legality.

FIX BEFORE SIGNOFF

Scope the signoff statement to "protocol-header transaction level"; list behavioural layers as explicit non-goals.

Required before the word "signoff" is used for design intent

#	Gate	Why it changes the conclusion
1	Bind a real RTL DUT (or fault-injected behavioural reference)	Turns a model self-test into design verification (RT-001).
2	Mutation testing with reported kill-rate	Proves the checkers can fail — the single most important missing number (RT-003).
3	Independent golden / spec-sourced vectors	Breaks the generator=checker circularity (RT-002, RT-004).
4	Non-vacuous SVA + vacuity covers	Makes assertions actually check behaviour (RT-005).
5	Pin CUS revision, diff model vs clause	Establishes model-to-spec fidelity (RT-008).
6	Commercial-sim run + real covergroup %	Simulator independence and true coverage (RT-006, RT-009).

Evidence base (all file:line-verified this audit): components/<slug>/rtl/* codec.sv components/<slug>/tb/*_tb_top.sv gen/gen_stim.py regression/code_cov.json regression/suite_regression_full.json lane2/ral, lane2/lane2b_native, ecpri_if.sv Verdicts: RT-001..010 CONFIRMED against source. This red-team does not retract the 1523/1523 result — it bounds its meaning. · AVIK VIP FACTORY — O-RAN VIP Suite.